

Manufacturing Cell Simulation — Biweekly Update

SimEvents Discrete-Event Model | Report date: 16 Sept 2026

Group members: Sayandip Mallick (2023MEB017) · Soumyajit Chakraborty (2023MEB015) · Chitrayan Biswas (2023MEB019) · Nilesh Banik (2023MEB003)

What we worked on this fortnight

The layout was already built, but it was running on fixed, made-up numbers and the QC rework loop was sitting there doing nothing. Over the last two weeks we made the model actually behave like a real, uncertain process. Four changes went in, and every one of them is now saved inside the model itself, not just in a script:

- **Random service times** everywhere instead of fixed ones — trucks, CNC machines, sub-assembly, QC checks, final assembly and packaging now all draw their processing time from a distribution each cycle.
- **Built-in SimEvents statistics turned on** at every server — utilization, average wait, and entities departed — with a scope hooked to each one so we can actually see how a station behaves instead of guessing.
- **QC reject/rework loop switched on.** The inspection blocks were already sitting in the layout, but nothing was driving the pass/fail decision. We gave each check its own attribute so CHECK I, CHECK II and CHECK III each make an independent call.
- **A basic breakdown proxy** on the three CNC machines — a small share of parts now trigger a long stoppage instead of the normal cycle time, standing in for a real MTBF/MTTR model until we get failure data.

One correction from last time: we'd said the three CNC machines were shared across all three raw-material lines. Looking at the actual model wiring, that's not right — each line has its own dedicated CNC machine (RM1 → Lathe, RM2 → Milling, RM3 → Shaping) running fully in parallel. The three lines only meet right at the end, after quality control, where they merge and get batched together for final assembly. The diagram on the next page shows this properly.

We ran the updated model for 20,000 minutes. Pages 2 and 3 show the model layout and the QC logic pulled straight out of the Simulink file, page 4 lists the exact parameter values on every block, and page 5 has the results.

Model layout — full process flow

Three raw-material lines run fully in parallel, each with its own truck unloading, CNC machine, sub-assembly and QC stage. They only merge right at the end, before final assembly. The view below is the model itself, scope taps included.

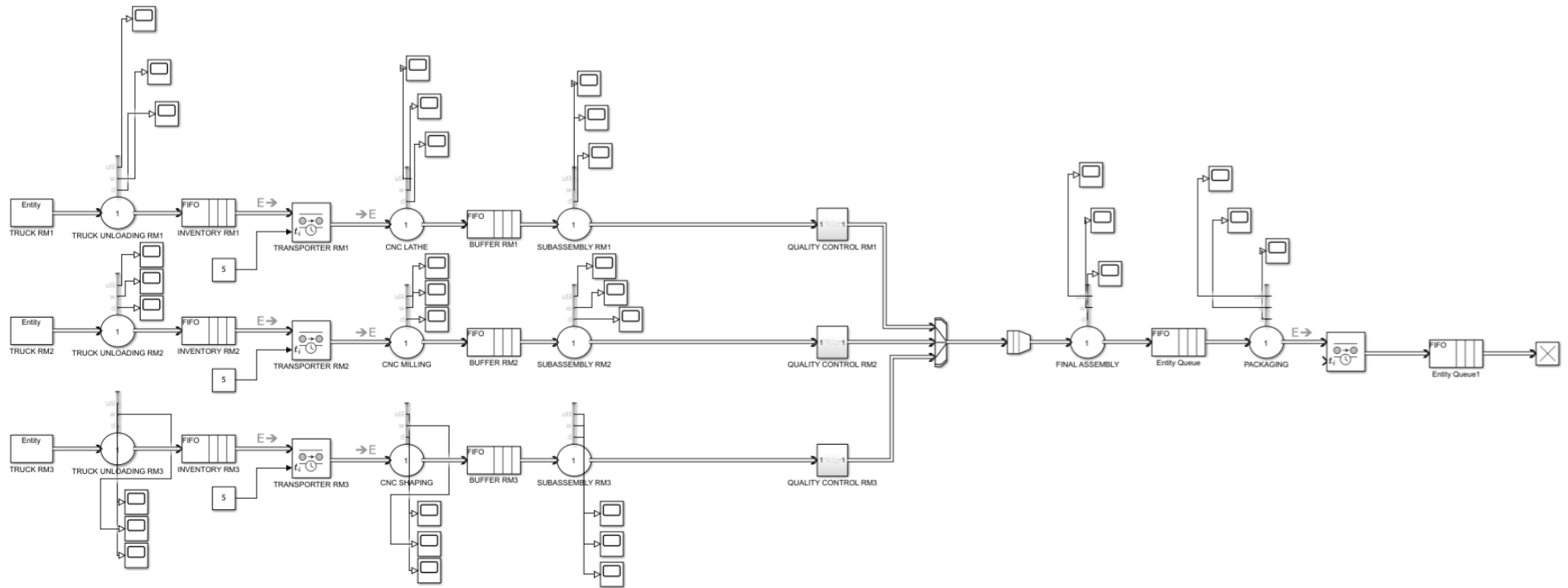


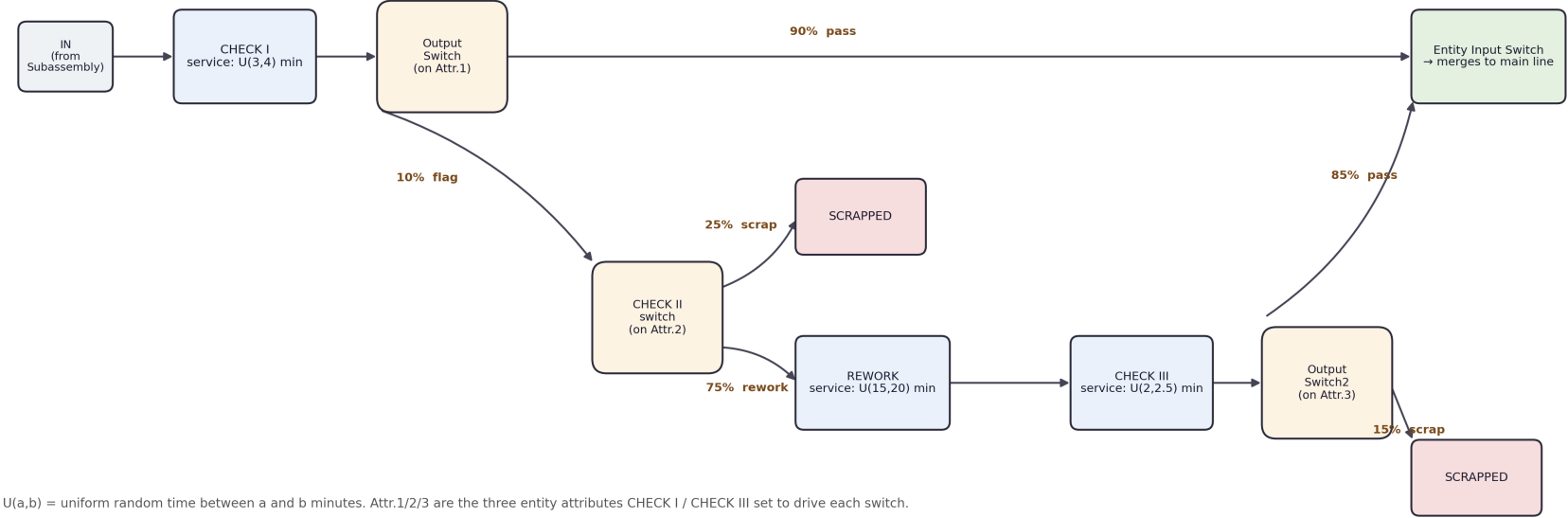
Fig. 1 — Manufacturing cell layout, captured directly from the Simulink/SimEvents model (three parallel RM lines, merging at the entity input switch before final assembly)

Each server carries three scope taps recording utilization, average wait and entities departed — see the appendix for the traces.

Zooming into the QC block — this is the part we just switched on

Each QUALITY CONTROL RM· block in Fig. 1 hides this same loop inside it. A part gets checked, 90% sail through, the rest get a second look and either get scrapped or reworked and re-checked. This is exactly the logic the upgrade script wires up, and it's what's now pulling parts out of the line before final assembly.

Quality Control loop — identical for RM1 / RM2 / RM3 (subsystem inside each QUALITY CONTROL RM· block)



U(a,b) = uniform random time between a and b minutes. Attr.1/2/3 are the three entity attributes CHECK I / CHECK III set to drive each switch.

Fig. 2a — QC reject/rework logic, identical inside all three QUALITY CONTROL blocks (percentages and timings read straight off the model's own switch/server settings)

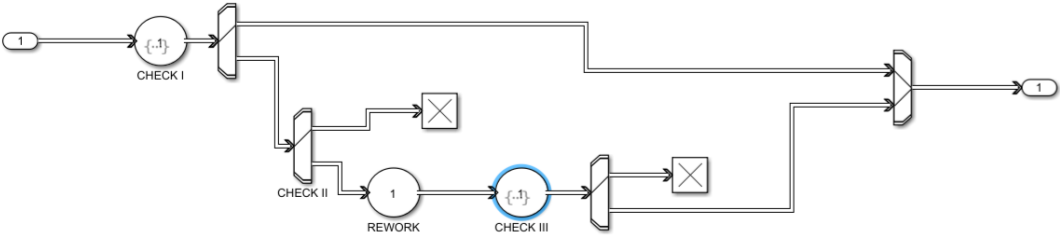


Fig. 2b — The same loop as it is actually wired inside the QUALITY CONTROL subsystem in the model: CHECK I → output switch → CHECK II → REWORK → CHECK III, with the two scrap terminators.

Parameters — what's actually set on each block

Pulled straight from the saved model (not just the script), so this is exactly what's running when you press play. dt is the service time in minutes; rand(1,1) is uniform 0-1, randn(1,1) is standard normal.

1. Service-time distribution on every server

Block	Meaning	ServiceTimeAction (exact)
TRUCK UNLOADING RM1 / RM2 / RM3	Uniform 8-10 min	$dt = 8 + 2 * \text{rand}(1,1);$
CNC LATHE / MILLING / SHAPING	96% of parts: Erlang(9,2), mean 18 min. 4% of parts: breakdown, uniform 60-90 min stoppage.	<pre>if rand(1,1) < 0.04 dt = 60 + 30*rand(1,1); else dt = -2*sum(log(rand(9,1))); end</pre>
SUBASSEMBLY RM1 / RM2 / RM3	Normal, mean 12, sd 3 (floored at 1 min)	$dt = \max(12 + 3 * \text{randn}(1,1), 1);$
QC — CHECK I	Uniform 3-4 min	$dt = 3 + 1 * \text{rand}(1,1);$
QC — CHECK III	Uniform 2-2.5 min	$dt = 2 + 0.5 * \text{rand}(1,1);$
QC — REWORK	Uniform 15-20 min	$dt = 15 + 5 * \text{rand}(1,1);$
FINAL ASSEMBLY	Normal, mean 20, sd 4 (floored at 1 min)	$dt = \max(20 + 4 * \text{randn}(1,1), 1);$
PACKAGING	Uniform 5-6 min	$dt = 5 + 1 * \text{rand}(1,1);$

2. QC pass / scrap / rework split (see Fig. 2)

Attribute	Set at	Split	Drives
Attribute1	CHECK I	90% → 1 10% → 2	Output Switch: 1 = straight through, 2 = to CHECK II
Attribute2	CHECK I	25% → 1 75% → 2	CHECK II switch: 1 = scrap, 2 = REWORK (only matters if Attribute1 = 2)
Attribute3	CHECK III	15% → 1 85% → 2	Output Switch2: 1 = scrap, 2 = merge back to main line

Net effect: out of 100 parts entering QC, about 90 pass first go, roughly 2-3 get scrapped immediately, about 7 get reworked and re-checked, and of those, ~1 fails again. So call it ~96 parts out of 100 make it through QC, ~4 get scrapped somewhere along the way.

3. Merging the three lines + stats

Block	Setting	What it does
Entity Input Switch	3 input ports	Merges the QC RM1 / RM2 / RM3 outputs into one stream
Entity Batch Creator	Batch size = 3	Groups 3 entities into 1 before final assembly — one finished unit needs one part from each RM line
Every server (unloading, CNC, sub-assembly, final assembly, packaging)	NumberEntitiesDeparted, AverageWait, Utilization = on	Turns on the built-in counters that feed the scopes shown in the appendix

What the scopes are showing

We read these off the scope traces after the 20,000-minute run, so treat them as ballpark figures rather than exact values — good enough to spot trends for now.

Station	Utilization	Avg. wait time (min)	Entities out (20,000 min)
Truck Unloading RM1	~3 – 4%	~9.1	67
CNC Lathe (RM1)	~6.5 – 7%	~20	67
CNC Milling (RM2)	~6.5 – 7%	~20	67
CNC Shaping (RM3)	~6.5 – 7%	~20	67
Sub-assembly RM1	~4 – 5%	~12.6	67
Final Assembly	~6.5 – 7.5%	~21	63
Packaging	~1.5 – 1.8%	~5.5	63

A few things stood out

- Nothing is anywhere close to maxed out — utilization sits in single digits almost everywhere. So the bottleneck isn't machine capacity, it's more about how the wait times stack up along each line.
- CNC machining and final assembly have the longest average wait (~20-21 min) — the slowest links right now. Packaging is by far the quickest (~5.5 min). Since the three RM lines run in parallel and only meet at the batch creator before final assembly, whichever line is slowest on a given part effectively sets the pace for that batch of three.
- About 67 parts leave truck unloading, the CNC stage and sub-assembly on each line by the end of the run, but only ~63 make it out of final assembly and packaging. That gap is the QC scrap/rework loop actually doing its job now — roughly what the ~4% scrap rate in the QC logic would predict.

Appendix — scope traces used above

Raw scope output for every station, for reference (x-axis: simulation time in minutes, 0-20,000).

